

Integrating Structural and Functional Genomics to Explain Divergence in Copper Stress Tolerance in *S. cerevisiae*

Abstract

Distinct selective environments elicit notable phenotypic diversity in *Saccharomyces cerevisiae*, particularly regarding tolerance to stressors such as metal toxicity. Oxidative stress induced by toxic heavy metals like copper necessitates a swift, coordinated transcriptional response to maintain cellular integrity and ensure survival. Strain-specific variations in gene copy number and downstream expression intensity frequently govern survival under extreme environments, despite high conservation of major stress-response pathways. This study investigates the extent to which copy number variation (CNV) and subsequent transcriptional plasticity across five key copper stress-regulation genes (*CUP1*, *SOD1*, *HSP12*, *CTR1*, and *CRS5*) explain observed differences in fitness across four diverse *S. cerevisiae* isolates: Y12, YJM1389, YPS128, and YPS163. Integrating RNA-seq, CNV, and growth phenotyping data reveals how gene expression responses dictate growth fitness under 10 mM CuSO₄.

Introduction

Baker's yeast, *Saccharomyces cerevisiae*, displays striking phenotypic diversity in its ability to tolerate environmental stressors, including heavy metal toxicity and oxidative stress. This divergence is observed even among closely related wild and industrial isolates. Consequently, these strains exhibit pronounced variation in survival and stress response strategies. A critical challenge for yeast in both wild and industrial environments is oxidative stress induced by metals such as copper (Cu). Excess intracellular copper generates cytotoxic

reactive oxygen species (ROS), requiring a rapid, concerted transcriptional response to prevent structural and cellular damage.

A primary drive of genomic variation in *S. cerevisiae* is Copy Number Variation (CNV), particularly within loci paramount for environmental adaptation. The cellular defense against copper enlists a multi-faceted regulatory network involving several key effectors. Metal detoxification relies on metallothioneins, encoded by *CUP1* and *CRS5*, which chelate free cytosolic copper ions to prevent toxic cross reactivity. High copy number variation at the *CUP1* locus dramatically increased copper tolerance, promoting survival under elevated metal concentrations. *CTR1* encodes a plasma membrane copper transporter that undergoes strict transcriptional and post-translational downregulation during copper stress to limit ion influx. Antioxidant defense relies on *SOD1*-encoded superoxide dismutase to neutralize ROS created by Fenton-like reactions catalyzed by copper, while *HSP12* provides membrane stabilization to resist broader proteotoxic stress (Morano et al., 2011; Hull et al., 2017).

While the isolated mechanisms of these stress-response genes are known, how baseline copy number variations translate into strain-specific transcriptional plasticity and overall growth fitness remains incompletely mapped across diverse backgrounds. To evaluate whether copy number variation (CNV) across copper homeostasis and stress-response loci explains strain-specific fitness divergence, four distinct *S. cerevisiae* isolates were underlyzed under extreme copper toxicity (10 mM CuSO₄). This analysis assesses whether CNV across primary metallothioneins (*CUP1*, *CRS5*), baseline transporters (*CTR1*), and primary stress effectors (*SOD1*, *HSP12*) dictates isolate-specific transcriptional plasticity. Strain-specific copy number expansions in core homeostasis genes (particularly at the *CUP1* locus) were hypothesized to

directly correlate with the magnitude of transcriptional activation, ultimately dictating strain-specific differences in growth fitness under copper stress.

Materials and Methods

RNA-Sequencing Data Processing and Alignment

Raw single-end RNA-sequencing (RNA-seq) reads across 36 samples were retrieved from the NCBI Sequence Read Archive (SRA) using prefetch and fastq-dump from the SRA toolkit. The dataset includes public accessions corresponding to four *S. cerevisiae* isolates, Y12 (SRR14156703–SRR14156711), YJM1389 (SRR14156757–SRR14156765), YPS128 (SRR14156775–SRR14156783), and YPS163 (SRR14156784–SRR14156791).

Raw sequencing reads underwent adapter trimming and quality filtering in Trim Galore (v0.6.10), retaining reads with Phred quality scores ≥ 20 and post-trimming lengths ≥ 36 bp. The resulting high-quality trimmed reads were aligned to an indexed *S. cerevisiae* S288C reference genome using STAR, producing coordinate-sorted BAM files. Transcript quantification was then executed via featureCounts from the Subread package by assigning single-end reads to coding sequence (CDS) features groups by the parent identifier from the S288C genome annotation file (.gff). Although the raw data consisted of single-end libraries, read quantification pipeline scripts explicitly include the -p flag during counting, treating reads under paired-end evaluation parameters during feature assignment.

Copy Number Variation and Phenotypic Data Integration

Copy Number Variation (CNV) matrices were extracted from the 1002 Yeast Genomes Project dataset. Pangenomic open reading frame (ORF) copy number estimates were filtered for

five target loci: metallothioneins CUP1 (YGR078C-A) and CRS5 (YOR031W), high-affinity copper transporter CTR1 (YOL112W), and stress effectors SOD1 (YJR104C) and HSP12 (YEL024W).

Strain-specific phenotypic fitness metrics were integrated from the 1002 Yeast Genomes Project normalized phenotyping dataset. Relative fitness was quantified as the strain growth ratio under stress conditions relative to standard YPD medium at 30°C. Growth ratios corresponding specifically to acute copper toxicity (10 mM CuSO₄) were extracted for each isolate to correlate with genomic variables.

Statistical Modeling and Feature Selection

To identify functional co-regulation patterns, transcriptomic profiles and baseline CNV metrics were evaluated using bivariate scatter plots overlaid with linear regression models. Prior to heatmapping and Principal Component Analysis (PCA), a variance-based filtering step was applied across the dataset. Transcripts demonstrating zero variance across the analyzed isolate samples were removed to eliminate unvarying genes and ensure statistical stability during dimensionality reduction.

Results

Correlation Analysis of Gene Expression vs Copper Fitness

Initial evaluation of transcriptomic expression across the five target loci revealed distinct functional groupings based on expression variation. *CTR1*, *HSP12*, and *SOD1* displayed zero variance in normalized expression across all evaluated isolates, maintaining constant baseline

normalized values of approximately 1.0. This expression profile indicates that baseline transport and general oxidative stress mechanisms are highly conserved and do not serve as primary drivers of strain-specific fitness divergence under acute CuSO_4 stress.

The metallothionein loci *CUP1* and *CRS5* showed strong transcriptional plasticity that scaled with isolate fitness. Low-fitness isolates, including SACE_YCK and AVI, displayed suppressed *CUP1* transcript levels (0.50) compared to high-performing strains ACK and AKN. Expression of *CRS5* followed this same positive trend. Despite sharing elevated expression profiles for both *CUP1* and *CRS5*, *ACK* and *AKN* differed substantially in fitness, yielding growth ratios of 1.10 and 0.20, respectively. Although metallothionein upregulation remains critical for baseline tolerance under 10 mM CuSO_4 , divergent growth rates across isolates demonstrate that genomic loci besides *CUP1* and *CRS5* determine overall cellular survival.

Hierarchical Clustering and Metallothionein-Driven Divergence

Hierarchical clustering based on Z-score normalized gene expression profiles divided the isolates into two distinct functional groups corresponding to their copper tolerance strategies: a high-tolerance cluster consisting of ACK and AKN, and a low-tolerance cluster comprising SACE_YCK and AVI. The high-tolerance cluster displayed high *CUP1* induction (Z-score +0.87), indicating that elevated metallothionein expression is necessary for survival under high copper concentrations. SACE_YCK and AVI exhibited *CUP1* suppression (Z-score -0.87). SACE_YCK displayed the lowest overall fitness in the cohort, characterized by simultaneous downregulation of both *CUP1* and *CRS5* (Z-score -1.50).

Principal Component Analysis of Genomic Fitness Differentiators

Principal Component Analysis (PCA) was conducted on the filtered expression matrix to identify primary genomic contributors to phenotypic variance. Excluding non-varying features (*CTR1*, *SOD1*, and *HSP12*) ensured that dimensional coordinates reflected active expression differences. The model captured 100% of total cohort variance across the first two principal components, with Principal Component 1 (PC1) accounting for 78.87% of total variance. Loading vectors for metallothioneins *CUP1* and *CRS5* aligned along the negative axis of PC1, confirming these two loci as major drivers of strain clustering. Low-fitness isolates SACE_YCK and AVI separated from metallothionein vectors along PC1, reflecting reduced metallothionein transcript abundance.

Discussion and Conclusion

This study evaluated the extent to which copy number variation and downstream transcriptional plasticity across five targeted loci influence copper stress survival in diverse *Saccharomyces cerevisiae* isolates. Integrating transcriptomic and growth phenotyping datasets demonstrates that strain-specific tolerance under acute copper toxicity 10 mM CuSO₄ is primarily mediated by the expression intensity of metallothioneins (*CUP1* and *CRS5*), rather than the divergence in copper transport (*CTR1*) or superoxide neutralization pathways (*SOD1*, *HSP12*). Principal Component Analysis confirmed that over three-quarters of all biological variance (78.87%) was captured by metallothionein transcript levels, separating high-tolerance strains from sensitive isolates.

The constant expression profile observed across *CTR1*, *SOD1*, and *HSP12* highlights an important methodological constraint within automated read-counting pipelines. Since read

quantification was executed using paired-end evaluation parameters (-p) on single-end libraries, a substantial fraction of aligned reads remained unassigned during feature counting. This uniform drop in assigned fragment density reduced transcript variance across unvarying loci, resulting in static normalized baselines (~ 1.0). While metallothionein loci (*CUP1* and *CRS5*) retained sufficient signal to demonstrate relative fitness divergence, these results emphasize the necessity of strict parameter matching during feature quantification to prevent systemic errors from obscuring underlying biological variation.

These findings highlight the evolutionary importance of pangenomic structural variation in driving ecological adaptation. Even among closely related yeast lineages, copy number expansions at specific locus hotspots produce distinct phenotypic divergence in extreme environments. Furthermore, the observation that isolate ACK significantly outperformed AKN despite nearly identical metallothionein expression profiles indicates that metallothionein amplification represents a necessary, but insufficient, requirement for maximum fitness. Future investigations expanding beyond these primary target ORFs will be essential to map additional non-coding regulatory elements and secondary metabolic pathways that adjust copper stress adaptation across the *S. cerevisiae* pangenome.

References

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